MILITARY SPECIFICATION

OSCILLATOR, HIGH VOLTAGE: 10539709

1. SCOPE

1.1 This specification covers a transitorized oscillator module. When energized by a 16.00 ± .20 volt direct current, the oscillator generates a sine wave voltage between 2.55 and 2.80 kilovolts peak-to-peak at a frequency between 1.2 and 2.0 kilocycles per second, and can deliver an output current of 10 milliamperes peak.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment

DRAWINGS

U.S. Army, Frankford Arsenal

D10539709	Oscillator, High Voltage: 10539709

FSC 1240

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PACKAGING DATA SHEET

MIL-P-14232/P10539709 Packaging of Oscillator, High Voltage

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The oscillator shall be manufactured in accordance with Drawing D10539709 and drawings pertaining thereto.

3.2.1 Moisture-fungus resistance.- The oscillator shall be treated per Type I for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances.
(b) Inorganic protective surface finishes.
(c) Part identification and marking.
(d) Workmanship.
(e) Soldering electrical connections.

3.4 Environmental service conditions.

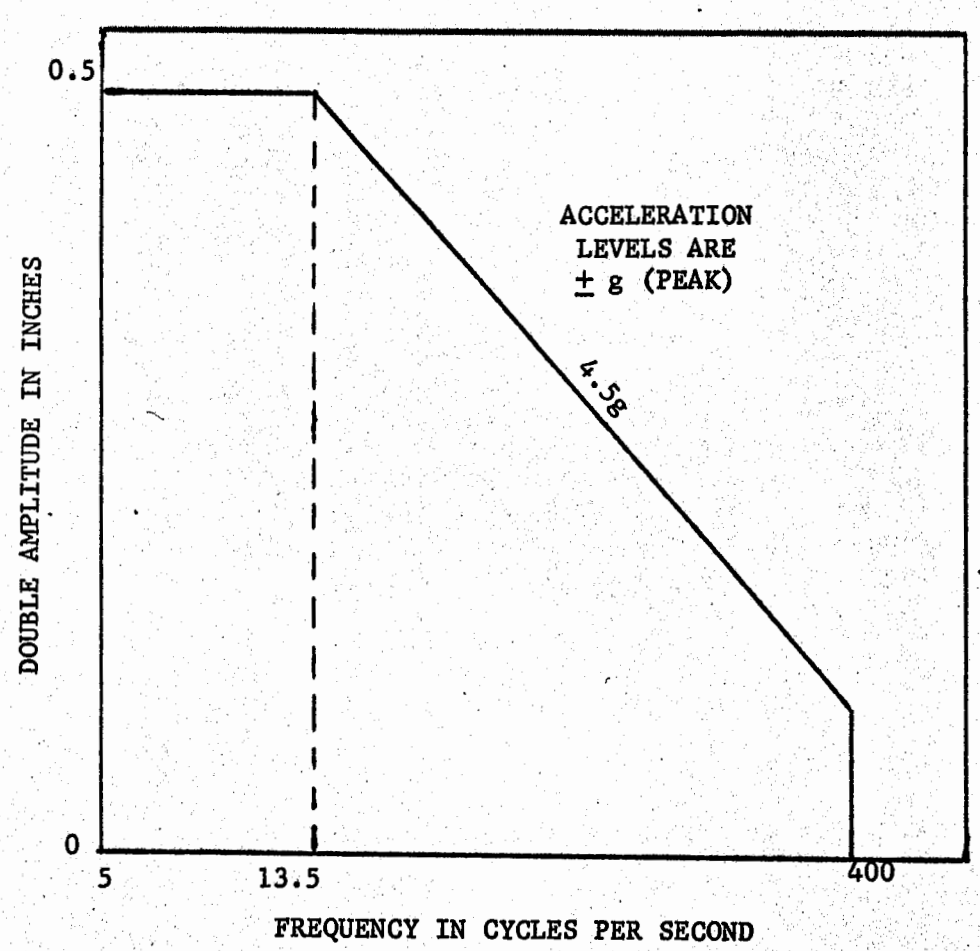
3.4.1 Vibration.- The oscillator shall show no evidence of physical failure due to being subjected to the following conditions:

a. Sinusoidal vibrations applied at the frequencies and amplitudes shown on Figure 1.
b. The applied frequency changed from 5 to 400 to 5 cycles per second (cps) at a logarithmic rate over a period of 15 ± 1.0 minutes.
c. A 30-minute dwell at each, not exceeding four, of the most significant resonant frequencies, if any.
d. The vibrations applied for three hours perpendicular to and three hours parallel to its longitudinal axes (total six hours).

During vibration, the oscillator shall exhibit an output voltage of 2700 +100 volts (V) peak-to-peak when operating in accordance with 3.5 and subjected to a capacitive load of 100 picofarads (pf). Subsequent to vibration the oscillator shall meet the requirements of 3.5.

3.4.2 Shock.-The oscillator shall withstand 3 shock impulses perpendicular to its longitudinal axes. The acceleration of each shock impulse shall be 300 gravity units (g). The nominal value for pulse deviation shall be 3 milliseconds. This requirement shall be met while being subjected to the following conditions:

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VIBRATION CURVE

FIGURE 1

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a. Input in accordance with 3.5.
b. Capacitive load of 100 pf.
c. Output voltage 2700 $^{+100}_{-150}$ V peak to peak

Subsequent to the shock impulses the oscillator shall show no evidence of physical failure and shall meet all the requirements of 3.5.

3.4.3 Operating temperatures.- The oscillator shall be capable of meeting the requirements of 3.5 when thermally stabilized at temperatures ranging from -65° to +150° Fahrenheit (F).

3.4.4 Storage temperatures.- The oscillator shall show no evidence of physical damage or electrical failure due to storage at temperatures ranging from -80° to +160°F. Subsequently the oscillator shall meet the requirements of 3.5 at a temperature between 60° and 90°F.

3.4.5 Humidity.

3.4.5.1 Operating.- The oscillator shall meet the requirements of 3.5, when subjected to atmospheres with a relative humidity (RH) up to 95% at a temperature of 150°F.

3.4.5.2 Non-operating.- The oscillator shall show no evidence of physical damage or electrical failure due to exposures in atmospheres of 100% RH, including condensation.

3.5 Performance.- Except as specified in 3.5.5, all the requirements shall be met with an input voltage of 16.0 ± 0.2 volts, direct current (vdc) at a current not to exceed 90 milliamperes.

3.5.1 Output voltages.- The output voltages of the oscillator shall be within the limits specified in Table I, when the oscillator is subjected to the respective load conditions.

TABLE I

Capacitive load Picofarads (pf)	Output voltages peak-to-peak
0	2700 ± 100
50	2700 ± 100
100	2700 $^{+100}_{-150}$

3.5.2 Output frequency.- With a load of 100 pf, the output frequency of the oscillator shall be between 1.3 to 1.8 kilocycles (kc). The frequency variation shall not exceed 10% when the capacitive load is removed.

3.5.3 Output current.- With a load of 100 pf, the oscillator's steady state output current shall not exceed 10 milliamperes peak.

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3.5.4 Short circuit.- The oscillator shall show no evidence of damage due to a direct short circuit applied between the ground terminal and the output terminal for a minimum of five minutes, while the oscillator is operating or starting.

3.5.5 Current surge and high voltage breakdown.- The dc current shall not exceed 120 milliamperes with no evidence of physical damage or electrical failure when subjected for a minimum of 10 minutes to the following conditions:

a. 17.00 vdc
b. 100 pf
c. 150°F

3.6 Reliability.- The oscillator shall exhibit a mean-time-between-failure (MTBF) of 1,500 hours with a confidence level of 90 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production

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testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables II, III, and IV herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.0% Defective		
101.	Output voltages	3.5.1	4.6.8.1
102.	Output frequency	3.5.2	4.6.8.2
103.	Output current	3.5.3	4.6.8.3
104.	Short circuit	3.5.4	4.6.8.4
105.	Current surge and high voltage breakdown	3.5.5	4.6.8.5
MINOR:	NONE		

NOTE: Unless otherwise specified, all tests shall be conducted at standard ambient temperature (60° to 90°F).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective

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units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One oscillator shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable drawings-visual
302.	Moisture-fungus resistance	3.2.1	4.6.1
303.	General specification	3.3	MIL-F-13926-visual and tactile

4.4.2 Environmental.

4.4.2.1 Extreme temperatures.- Three oscillators shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each oscillator shall meet the requirements and tests in Table IV.

TABLE IV

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (non-operating)	3.4.4	4.6.2
305.	Humidity (non-operating)	3.4.5.2	4.6.3
306.	Temperature (operating)	3.4.3	4.6.6
307.	Humidity (operating)	3.4.5.1	4.6.7

4.4.2.2 Shock and vibration.- At 25%, 50%, and 75% completion of the total number of oscillators under a production contract, 5 oscillators shall be selected at random from each representative interval of production and shall meet the requirements and tests in Table V.

TABLE V

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
308.	Vibration	3.4.1	4.6.4
309.	Shock	3.4.2	4.6.5

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4.4.3 Failure of a sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.1 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and shall be connected as shown in Figure 2.

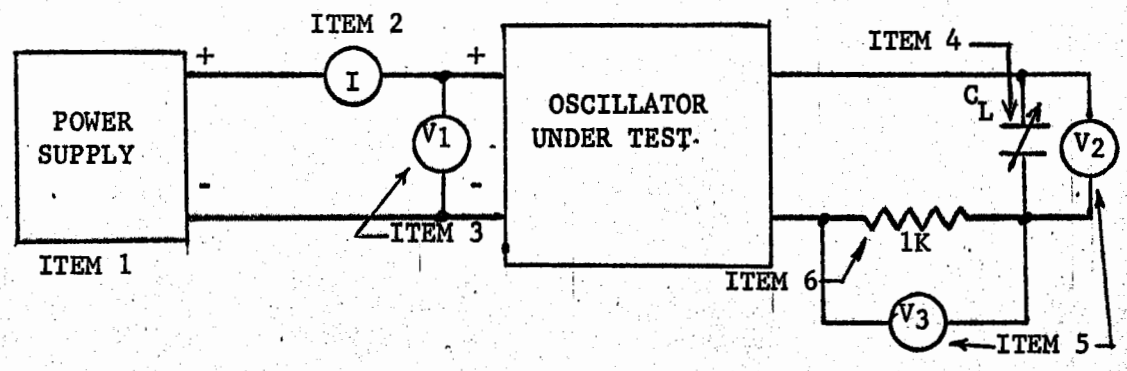


FIGURE 2

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ITEM	DESCRIPTION
1. Power Supply, d.c.	1% regulation, 100 mv ripple, to supply input voltages as specified in 3.5 and 3.5.6.
2. Ammeter, d.c.	Accuracy ± 3% of reading, to measure input current as specified in 3.5 and 3.5.6.
3. Voltmeter, d.c.	Accuracy ± 0.01% of reading, to measure input voltages as specified in 3.5 and 3.5.6.
4. Capacitor	Total capacitance of 50 and 100 picofarads (includes any capacitance of external components), ± 1%.
5. Oscilloscope	Accuracy ± 1% of reading (peak to peak), to verify output current, output voltage and frequency, or other suitable means.
6. Resistor	1000 ohms, 1/4 watt (minimum), ± 0.1% tolerance.

4.6 Test methods and procedures.

4.6.1 Moisture-fungus resistance.- The oscillator shall be submitted for inspection after treatment as specified in 3.2.1, to determine compliance with the inspection provisions of MIL-T-152.

4.6.2 Storage temperatures.- This test shall utilize test equipment conforming with the "Test facilities" requirement of MIL-F-13926 and the conditions specified in requirement 3.4.4. Temperatures shall be gradually changed in the performance of this test as a precautionary measure to avoid the possibility of thermal shock. After oscillator temperature stabilization at - 80°F and +160°F, the temperature shall be maintained constant for 24 hours at each temperature specified. The oscillator shall subsequently meet requirements of 3.5 (between 60°F and 90°F) to determine compliance with the requirement 3.4.4.

4.6.3 Humidity non-operating condition.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the relative humidity shall be the value specified in 3.4.5.2 for not less than four hours and the test shall cease at the end of that period. Upon return to a temperature of between 60°F and 90°F the external surfaces of the oscillator shall be thoroughly dried by wiping all accessible areas. Drying by air blast or in an oven shall not be permitted. After a period of not more than

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two hours the oscillator shall be visually inspected for evidence of physical damage and shall then meet requirements of 3.5 to determine compliance with requirement 3.4.5.2.

4.6.4 Vibration-operating.- This test shall utilize test equipment specified in 4.5.3 in addition to standard vibrating equipment with means for mounting the oscillator and controlling the direction of vibration to assure compliance with 3.4.1. Vibration equipment shall provide a means of measuring amplitude and frequencies of vibration to comply with the limits specified in 3.4.1. Test equipment measuring instruments employed shall measure vibration amplitude within + 5% and vibration frequency within + 2%. Vibration shall be under conditions specified in the "Test facilities" of MIL-F-13926 with the oscillator energized as specified in 3.5. During this test a total capacitance of 100 picofarads shall be across the output terminals of the oscillator. To comply with requirement 3.4.1, observe across the oscillator's output that the alternating current (ac) output peak to peak voltage remains at a level within 2700 + 100 volts, with no evidence of oscillator intermittent operation or high voltage breakdown during this test, or when subsequently meeting the requirements of 3.5.

4.6.5 Shock-operating.- This test shall utilize test equipment specified in 4.5.3 and shall be conducted under conditions specified in the "Test facilities" requirement of MIL-F-13926. This test shall be in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the number and direction of applied shocks shall be as specified in 3.4.2. The oscillator shall be energized as specified in 3.5 and has a total capacitance of 100 picofarads across its output terminals. To comply with requirement 3.4.2 observe that oscillator output voltage across it's output terminals remains at a level within 2700 + 100 vac with no evidence of oscillator intermittent operation or high voltage breakdown during this test, or when subsequently meeting the requirements of 3.5.

4.6.6 Temperature-operating.- This test shall utilize test equipment specified in 4.5.3 in addition to environmental test equipment conforming with the "Test facilities" requirement of MIL-F-13926. Test facilities shall maintain conditions specified herein in accordance with requirement 3.4.3. Temperatures shall be gradually changed in the performance of this test as a precautionary measure to avoid the possibility of thermal shock. After the oscillator has attained thermal stabilization at -65°F and then at +150°F, temperature shall be maintained constant at each temperature while determining compliance with all requirements of 3.5. To determine compliance with requirement 3.4.3, inspect the oscillator under conditions specified in the individual electrical test specified in 4.6.9 for meeting the associated requirement of 3.5 and for not less than one hour at each extreme temperature specified in 3.4.3.

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4.6.7 Humidity-operating.- This test shall utilize test equipment specified in 4.5.3. The test shall be performed in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that temperature and humidity shall be maintained at the value specified in requirement 3.4.5.1 for a 24 hour period and the test shall cease at the end of that period. To determine compliance with requirement 3.4.5.1 observe that the oscillator meets all requirements of 3.5 just prior to completion of this test without evidence of oscillator intermittent operation or high voltage breakdown.

4.6.8 Performance of electrical tests.- Electrical tests specified herein shall be performed under a maximum relative humidity of 80%, a barometric pressure of 28 to 30 inches of mercury, at a temperature between 60°F and 90°F, except where otherwise specified for conditions in requirement 3.4.3, 3.4.5.1 and 3.5.5.

4.6.8.1 Output voltages.- Energize the oscillator as specified in 3.5 for not less than 15 minutes, then perform this test. Utilize test equipment items in 4.5.3 as appropriate, in the test circuit in Figure 2. To determine compliance with 3.5.1 establish in sequence each output load condition specified in Table 1 and verify oscillator peak to peak output voltages under each load condition.

4.6.8.2 Output frequency.- Energize the oscillator as specified in 3.5 with a total capacitance load of 100 pf. Observe that the oscillator output frequency is not less than 1.3 kilocycles per second and not more than 1.8 kilocycles per second. With the output load capacitor removed and a capacitive load due to external components not greater than 5 pf the output frequency variation shall not exceed 10%.

4.6.8.3 Output current.- Energize the oscillator as specified in 3.5 with a total capacitance load of 100 pf and observe that the oscillator steady state output current is not more than 10 milliamperes peak.

4.6.8.4 Short-circuit.- Energize the oscillator as specified in 3.5. Operate the oscillator with no resistive or reactive load connected across it's output terminals. To determine compliance with requirement 3.5.4 connect (short-circuit) oscillator output terminals with an electrical conductor having a dc resistance of not more than 0.1 ohm for not less than five minutes. Remove the short-circuiting conductor and observe that no evidence of oscillator physical damage or deterioration in oscillator performance exists.

4.6.8.5 Current surge and high voltage breakdown.- Energize the oscillator as specified in 3.5.5 under the conditions therein and observe that the dc current shall not exceed 120 milliamperes with no evidence of physical damage or electrical failure.

4.6.9 Reliability.- Reliability shall be in accordance with Test Plan A of MIL-STD-781.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10539709. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- When used with the next higher assembly the oscillator is primarily intended to supply the electrical power requirements of the image intensifier tube assembly (25 Millimeter) Type 8586, employed in Tank Periscopes XM50 and XM51.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Order of document precedence.
c. Selection of applicable levels of preservation, packing, and packing.
d. Applicable packaging data sheet (see 5.1).

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Preparing activity:

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